KAIZEN MATHS

# Lesson starter

**Binomial and Geometric Distributions**

Mixed & Inverse

Name

**S1**

A player repeats a game until they win. The probability of winning each game is 0.45. Which distribution should model the trial of the first win, and what probability would find the first win on trial 6?

**S2**

X ~ B(6, p). Given p = 0.6, calculate P(X ≥ 1).

**S3**

A test has 15 independent questions. The probability of answering each correctly is 0.75. Which distribution should model the number correct, and what probability would find exactly 2 correct?

**S4**

X ~ B(8, p). Given p = 0.5, calculate P(X ≥ 1).



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Show your working

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# Exit ticket

**Binomial and Geometric Distributions**

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Name _____

**EA1**

A test has 8 independent questions. The probability of answering each correctly is 0.6. Which distribution should model the number correct, and what probability would find exactly 6 correct?

**EA2**

X ~ B(11, p). Given p = 0.25, calculate P(X ≥ 3).



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Exit ticket

Binomial and Geometric Distributions

Mixed & Inverse

Name _____

**EB1**

A test has 10 independent questions. The probability of answering each correctly is 0.75. Which distribution should model the number correct, and what probability would find exactly 5 correct?



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Name _____

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A test has 8 independent questions. The probability of answering each correctly is 0.6. Which distribution should model the number correct, and what probability would find exactly 6 correct?

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# Teacher answers & working

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**S1**

A player repeats a game until they win. The probability of winning each game is 0.45. Which distribution should model the trial of the first win, and what probability would find the first win on trial 6?

**Answer:**

Geometric, X ~ Geo(0.45)

**Working:**

The process continues until the first success.

Use a geometric distribution: X ~ Geo(0.45).

First win on trial 6 would be $P(X = 6) = (0.55)^5(0.45)$.

Geometric

**S2**

X ~ B(6, p). Given p = 0.6, calculate $P(X \geq 1)$.

**Answer:**

0.9959

**Working:**

Use the complement for an at-least probability.

$P(X \geq 1) = 1 - P(X \leq 0)$.

With p = 0.6, $P(X \leq 0) \approx 0.0041$.

$P(X \geq 1) \approx 0.9959$.

0.9959

**S3**

A test has 15 independent questions. The probability of answering each correctly is 0.75. Which distribution should model the number correct, and what probability would find exactly 2 correct?

**Answer:**

Binomial, X ~ B(15, 0.75)

**Working:**

The number of trials is fixed and each trial has success/failure.

Use a binomial distribution: X ~ B(15, 0.75).

Exactly 2 correct would be $P(X = 2) = 15C2(0.75)^2(0.25)^{13}$.

Binomial

# Teacher answers & working

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**S4**

X ~ B(8, p). Given p = 0.5, calculate P(X ≥ 1).

**Answer:**

0.9961

**Working:**

Use the complement for an at-least probability.

$P(X \geq 1) = 1 - P(X \leq 0)$.

With p = 0.5, P(X ≤ 0) ≈ 0.0039.

$P(X \geq 1) \approx 0.9961$.

0.9961

**EA1**

A test has 8 independent questions. The probability of answering each correctly is 0.6. Which distribution should model the number correct, and what probability would find exactly 6 correct?

**Answer:**

Binomial, X ~ B(8, 0.6)

**Working:**

The number of trials is fixed and each trial has success/failure.

Use a binomial distribution: X ~ B(8, 0.6).

Exactly 6 correct would be $P(X = 6) = 8C6(0.6)^6(0.4)^2$.

Binomial

**EA2**

X ~ B(11, p). Given p = 0.25, calculate P(X ≥ 3).

**Answer:**

0.5448

**Working:**

Use the complement for an at-least probability.

$P(X \geq 3) = 1 - P(X \leq 2)$.

With p = 0.25, P(X ≤ 2) ≈ 0.4552.

$P(X \geq 3) \approx 0.5448$.

0.5448

# Teacher answers & working

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**EB1**

A test has 10 independent questions. The probability of answering each correctly is 0.75. Which distribution should model the number correct, and what probability would find exactly 5 correct?

**Answer:**

Binomial, $X \sim B(10, 0.75)$

**Working:**

The number of trials is fixed and each trial has success/failure.

Use a binomial distribution: $X \sim B(10, 0.75)$.

Exactly 5 correct would be $P(X = 5) = 10C5(0.75)^5(0.25)^5$.

Binomial